

Aditya Singh

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EDUCATION

Scaler School Of Technology

Bachelor of Engineering in Computer Science | CGPA: 7.1 / 10.0

Bengaluru, Karnataka

July 2023 – July 2027

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Computer Science | CGPA: 8.3 / 10.0

Pilani, Rajasthan

July 2023 – July 2027

EXPERIENCE

1Pharmacy

AI/ML & Fullstack Intern (PPO: 30 LPA)

Bengaluru, India

April 2026 – Present

- Architected a real-time person identification pipeline combining Facial Recognition with a Vision Language Model (VLM): annotated live RTSP video frames with detected faces, bounding boxes, and identity metadata, then streamed enriched context to the VLM for natural-language scene understanding and person-level Q&A over multi-person feeds.
- Built an end-to-end staff attendance and dwell-time analytics system on multi-camera CCTV feeds: automated check-in/check-out via face recognition, zone-wise dwell-time computation, and per-employee presence reports, replacing manual attendance entirely.
- Solved Person Re-Identification (ReID) under a uniform-clothing constraint — since all staff wear identical uniforms, appearance-based ReID embeddings failed; engineered a hybrid approach fusing face-anchored identity assignment, body-embedding similarity, and spatio-temporal track association (motion continuity + camera topology) for robust cross-camera multi-object tracking.
- Designed a hybrid edge-cloud backend: on-edge inference (frame ingestion, face/person detection, embedding extraction) with cloud-side aggregation, storage, and analytics REST APIs — cutting bandwidth usage and end-to-end identification latency.
- Took full ownership of the Sanjai internal platform, driving architecture decisions, database schema design, backend APIs, and frontend delivery end-to-end; PPO extended within the internship based on performance and ownership.

Aaritya — sahi.com

Mobile Developer Intern

Bengaluru, India

Sept. 2024 – June 2025

- Developed the complete public-facing showcase website for sahi.com using Next.js, TypeScript, and TailwindCSS with a fully reusable component architecture.
- Optimised performance via dynamic imports, compressed assets, and improved image loading, achieving measurable Lighthouse performance score improvements.
- Contributed to the internal trading application built in Flutter (Dart), shipping reusable widget architecture, state management improvements, and UI enhancements across core trading flows.

Dukaan

Backend Development Intern

Bengaluru, India

July 2024 – Sept. 2024

- Built and improved backend services using Node.js, Express.js, and MongoDB for internal operations and business workflows.
- Implemented secure JWT authentication with refresh token rotation and a team-based multi-session login architecture supporting concurrent user access.

PROJECTS

rnassist | Node.js, WebSocket, LLM Integration, npm | 600+ downloads

2024 – Present

- Built and published a developer CLI tool that hooks into Metro's WebSocket debugger to capture runtime errors, console logs, and network calls in real time with zero app-side instrumentation required.
- Scans project `package.json` against a known-bug database to flag vulnerable dependency versions with actionable fix suggestions surfaced directly in the terminal.
- Sends collected runtime metrics and source files to an LLM for automated code review, surfacing performance issues and anti-patterns with file-level specificity.

Watchman (Open Source) | Python, OpenCV, ONNX Runtime, FastAPI, Docker, WebSockets

2026

- Building a self-hostable, plug-and-play edge video analytics toolkit: connect any RTSP camera and get person detection, tracking, ReID, and zone-based event alerts (entry, loitering, count) out of the box on low-cost hardware.
- Modular inference pipeline with swappable ONNX models, an event webhook system, and a live web dashboard streaming annotated frames over WebSockets; fully containerized with Docker for one-command deployment.

OneRoot | React Native, Supabase, Cloud Functions, YOLO | Live on Play Store

2024

- Delivered a production-ready B2B marketplace published on the Google Play Store enabling end-to-end trading workflows between farmers, vendors, and buyers.
- Built real-time admin approval workflows using Supabase Realtime channels for controlled listing visibility and dynamic state management.
- Integrated a YOLO machine learning pipeline for live load estimation directly from camera feed, enabling transparent and automated deal verification.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Dart, Java, SQL

AI/ML & Computer Vision: Vision Language Models (VLM), LLM Integration, Facial Recognition, Person Re-Identification (ReID), Multi-Object Tracking, YOLO, OpenCV, ONNX, Edge AI Inference, RTSP Video Pipelines

Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets, Socket.io, JWT/OAuth, Microservices, Supabase

Mobile & Web: React Native, Flutter, Next.js, React, Redux, TailwindCSS

Databases & Messaging: PostgreSQL, MongoDB, MySQL, Redis, Firebase

Cloud & DevOps: Docker, Git, GitHub Actions, CI/CD, AWS/GCP, Linux, Nginx

System Design: Edge-Cloud Architecture, DB Indexing, Sharding, Consistent Hashing, Caching, CDN, Load Balancing, LLD